

APP DEVELOPMENT - 2 PROJECT REPORT

1. Student Details

Name: Dilshaz Hussain

Roll Number: 24F3003988

Email: 24f3003988@ds.study.iitm.ac.in

About Me:

I am pursuing the BS Degree in Data Science and Applications at IIT Madras. I have a strong interest in full-stack web development and building real-world applications. This project enhanced my understanding of Vue.js, Flask REST APIs, SQLAlchemy ORM, SQLite, Redis, Celery, caching and asynchronous task processing.

2. Project Details

Project Title: Placement Portal Application (PPA) - Version 2

Problem Statement:

The objective of this project is to design and develop a web-based placement portal where students can apply for placement drives, companies can conduct recruitment, and the institute placement cell can efficiently manage approvals, applications, and placement activities.

Approach:

The application uses Vue.js for the frontend and Flask for backend APIs. SQLAlchemy ORM manages SQLite database operations. Redis and Celery are integrated for caching and asynchronous tasks such as CSV export, daily reminder emails, and monthly activity reports. The system follows secure role-based authentication for Admin, Company, and Student users.

3. AI / LLM Declaration

AI tools were used for:

- Debugging Flask, Vue.js and SQLAlchemy issues
- Understanding implementation concepts
- Improving user interface design
- Preparing documentation

The approximate usage of AI tools was 15–20%, mainly for guidance and debugging. All application logic, implementation, testing and validation were completed manually.

4. Technologies and Frameworks Used

Technology	Purpose
Vue.js 3 & Vue Router	Frontend
Flask	Backend APIs
SQLAlchemy	ORM
SQLite	Database
Bootstrap 5	Responsive UI
Redis	Caching
Celery	Background Jobs
Flask-Mail	Email
Flask-Caching	Performance

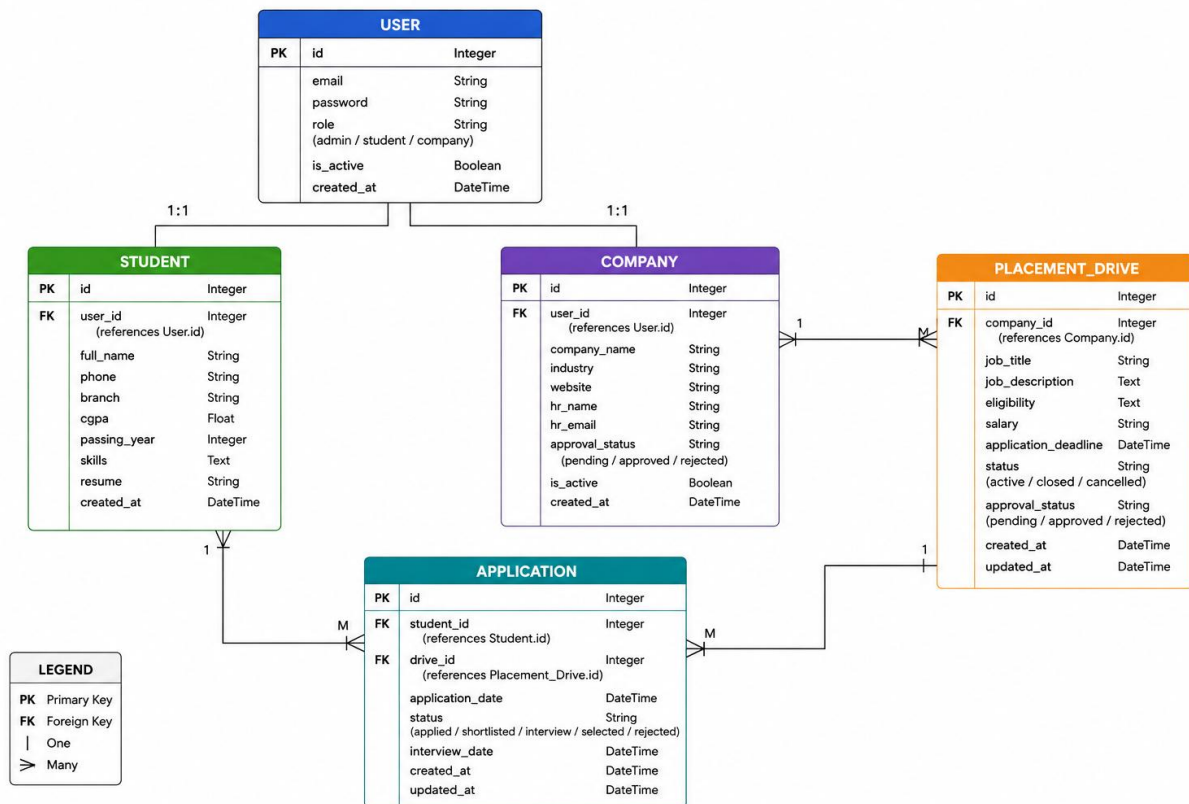
5. Database Schema Overview

Tables:

- User
- Student
- Company
- PlacementDrive
- Application

Relationships:

- One User → One Student
- One User → One Company
- One Company → Many Placement Drives
- One Student → Many Applications
- One Placement Drive → Many Applications



6. REST API Endpoints

Endpoint	Method	Description
/home	GET	Home page
/login	POST	User Login
/student/register	POST	Student Registration
/company/register	POST	Company Registration
/admin/dashboard	GET	Admin Dashboard
/student/drives	GET	View Drives
/student/apply/<id>	POST	Apply
/company/drive	POST	Create Drive
/company/applications	GET	Manage Applicants

7. Application Architecture and Features

Architecture:

Vue.js Frontend

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Flask REST APIs

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SQLAlchemy ORM

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SQLite Database

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Redis + Celery

Student:

- Registration & Login
- Profile Management
- Resume Upload
- Search Drives
- Apply for Drives
- Application Status
- CSV Export

Company:

- Registration
- Create Drives
- Manage Applicants
- Shortlist Students
- Interview Scheduling
- Final Selection

Admin:

- Dashboard
- Approve Companies
- Approve Drives
- Search
- Blacklist Users
- View Applications

Background Jobs:

- Daily Reminder Emails
- Monthly Activity Report
- Asynchronous CSV Export
- Caching

8. Video Presentation

Drive Link: <https://drive.google.com/file/d/15kCYHbcoq-XN-8DDpqHRdVQ9NgCgEX-/view?usp=sharing>

9. Conclusion

The Placement Portal Application (Version 2) automates the complete placement workflow using Vue.js, Flask, SQLAlchemy, SQLite, Redis and Celery. The application provides secure role-based access, background processing, responsive user interface and efficient placement management while fulfilling the MAD-2 course requirements.